1· Set up

(a) Attention weight. Each head in each layer gives the final token T a probability distribution over every earlier position, BOS included, so the weights sum to 1 and attention gained at one position is lost at others. a is the share of one head's lookup that lands on position τ.

(b) Intervention. A row is BOS plus 32 random tokens with no repeats. T (position 32) is the query and is never edited. For each $\tau = 1 \dots 31$, one of 11 countries replaces the token at τ and nothing else changes. The unedited row is the control, so any change in a is caused by the token at τ, through its key or through how earlier layers changed T's query.

control BOS | she | sc | … 27 random tokens … | age | dis | ious **country** BOS | she | sc | … 27 random tokens … | France | dis | ious (τ = 30, d = 2)

(c) The number. $\Delta = a_{\text{country at } \tau} - a_{\text{control at } \tau}$, for each head and each distance $d = 32 - \tau$ ($d = 1$ is the adjacent slot), averaged over the 11 countries so each row counts once.

(d) Scale. 200 rows per filler type (common: token ids 256–999, rare: 1000–39999) × 11 countries × 31 positions, 68,200 forward passes per filler type. The result is an array $x[\text{row}, d, \text{layer}, \text{head}]$ of size $200 \times 31 \times 12 \times 12$.

$$x[r, d, \ell, h] = \frac{1}{11} \sum_{c=1}^{11} a^{(c)}_{r,\ell,h}(\tau) \; - \; a^{0}_{r,\ell,h}(\tau) \; = \; \mathbb{E}_c\left[a^{(c)}_{r,\ell,h}(\tau)\right] - a^{0}_{r,\ell,h}(\tau), \qquad \tau = 32 - d$$

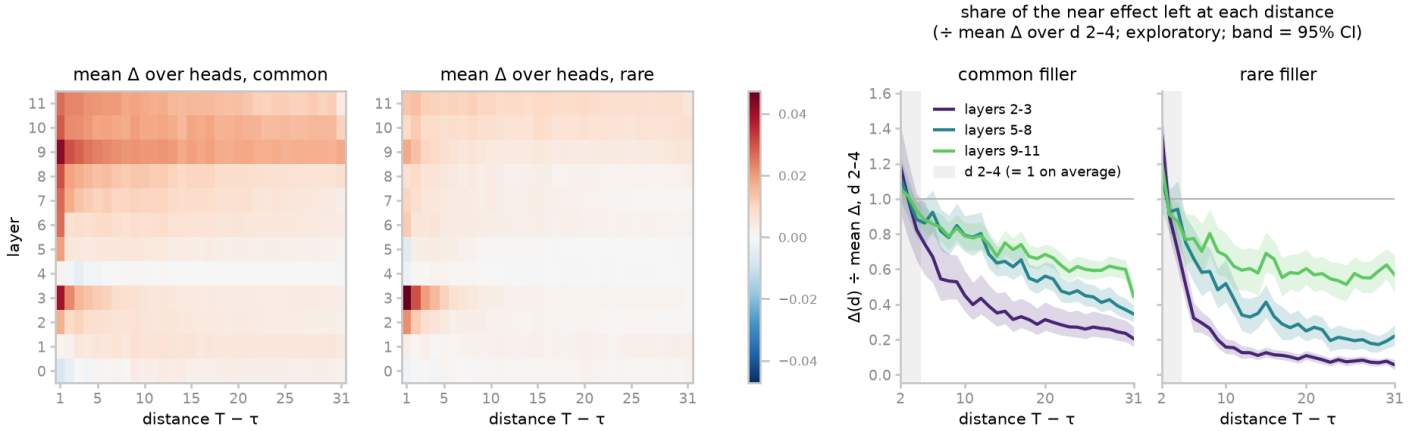
$$M_\ell(d) = \frac{1}{200} \sum_{r=1}^{200} \; \frac{1}{12} \sum_{h=1}^{12} x[r, d, \ell, h] \; = \; \mathbb{E}_{r,\,h}[x[r, d, \ell, h]]$$

Let

$r \in \{1, \dots, 200\}$	be the row index,
$c \in \{1, \dots, 11\}$	be the country index,
$\ell \in \{0, \dots, 11\}$	be the layer index,
$h \in \{1, \dots, 12\}$	be the head index,
$d \in \{1, \dots, 31\}$	be the distance,
$\tau = 32 - d$	be the position of the inserted country,
$a^{(c)}_{r,\ell,h}(\tau)$	be the attention from T to position τ in row r, layer ℓ, head h, with country c at τ,
$a^{0}_{r,\ell,h}(\tau)$	be the same quantity in the control row,
$\mathbb{E}_c$	be the uniform mean over the 11 countries,
$\mathbb{E}_r$	be the uniform mean over the 200 rows,
$\mathbb{E}_h$	be the uniform mean over the 12 heads of layer ℓ, and
$M_\ell(d)$	be the value of one heatmap cell (layer ℓ, distance d), computed separately for each filler type.

2· Figures

no model run: saved arrays only



- **Right.** Let $\mathcal{N} = \{2, 3, 4\}$ be the near distances ($d = 1$ is left out) and $G \in \{\{2, 3\}, \{5, \dots, 8\}, \{9, 10, 11\}\}$ a layer group. Each curve is, for $d = 2, \dots, 31$,

$$\text{share}_G(d) = \frac{\mathbb{E}_{\ell \in G}\, M_\ell(d)}{\mathbb{E}_{\ell \in G,\, d' \in \mathcal{N}}\, M_\ell(d')}$$

  - **Layers 2–3 (24 heads).** The steepest fall: 0.49 (common) and 0.22 (rare) of the near effect over d = 5–15, then 0.28 and 0.09 over d = 16–31.
  - **Layers 5–8 (48 heads).** 0.78 (common) and 0.49 (rare) over d = 5–15, then 0.48 and 0.23 over d = 16–31.
  - **Layers 9–11 (36 heads).** The flattest: 0.79 and 0.68 over d = 5–15, then 0.63 and 0.57 over d = 16–31. With common filler it is no different from layers 5–8 over d = 5–15 (0.79 vs 0.78).